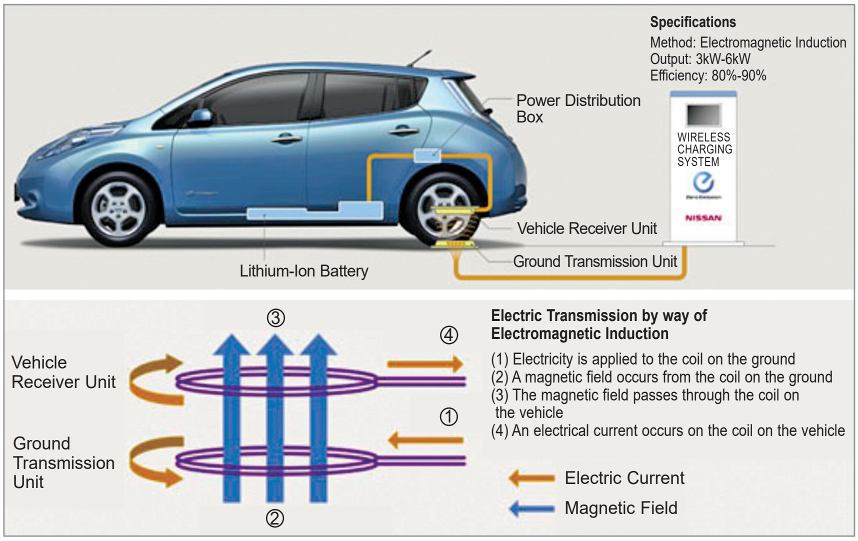
and development, 2.5D packaging technologies like TSMC's chip-on-wafer-on-substrate (CoWoS) platform, Intel's embedded multi-die interconnect bridge (EMIB) technology, and Samsung's I-Cube, which enhance AI computational power by integrating computing chips and memory, are now widely used in high-performance chips. TrendForce reports that by 2024, suppliers are set to focus on ramping up 2.5D packaging capacity to meet the rising demand for high computational power in applications like AI.

Meanwhile, we also see companies unveiling their 3D packaging technologies, like TSMC's system on integrated chips (SoIC) 3D fabric, Samsung's X-Cube, and Intel's Foveros. 3D packaging usually uses through-silicon vias (TSVs) to do away with the silicon interposer layer used in 2.5D.

Quantum computing is here. In a media story, Tony Uttley, President and Chief Operating Officer, Quantinuum, pointed out that quantum computers are not 10-15 years away. "We have quantum computers right now, which can do things that classical computers can't. Another misconception is that quantum computers will only be good for one thing," he had remarked.

He went on to explain that while the primary focus of quantum computing development has been on business applications, the technology has inadvertently become a game-changer for complex scientific challenges like condensed matter physics and high-energy physics problems

In October 2023, Atom Computing unveiled a 1,225-site atomic array, currently populated with 1180 qubits, in its next-generation quantum computing platform, which is scheduled for launch this year. The company claimed to be the first to cross the 1,000-qubit threshold for a universal gate-based system. Experts predict that in the coming year, the focus will shift from raw qubit count to logical qubit count, focusing on quality of qubits, error correction, and functional applications.



EV makers like Nissan are exploring wireless charging of EVs (Courtesy: Nissan)

In December 2023, IBM debuted the 133-qubit IBM Quantum Heron processor, the first in a new series of utility-scale quantum processors. Its high performance metrics and low error rates surpass all earlier IBM quantum processors

The company also unveiled the IBM Quantum System Two, the company's first modular quantum computer, and the foundation of IBM's quantum-centric supercomputing architecture. It combines scalable cryogenic infrastructure and classical runtime servers with modular qubit control electronics.

IBM is also pioneering the use of genAI for quantum code programming through watsonx. They hope that this will simplify the development of quantum algorithms for utility-scale exploration.

Software trends: Aye, it is AI again!

TuringBots to improve productivity across the SDLC. A report by Chris Gardner, VP and Research Director, Forrester, predicts that TuringBots will improve the software development lifecycle (SDLC) productivity by 15-20%. "2023 proved that generative AI has an exponential impact on software development. In 2024, many development teams will go from experimentation to embedding TuringBots in their software development lifecycle. Coders will gain 20-50% productivity on average— some have seen productivity rise to 200% and more when experienced engineers use genAI to work with languages or libraries outside their normal zone of development. Testers will also gain 15-20% productivity, and all members of product teams will gain above 10% efficiency from their assistive TuringBots in planning and delivery. GenAI will make low-code and high coding much more productive everywhere, and this will exponentially grow going forward," he writes.

Ajar source licensing ploys. Last year, Forrester researchers also brought to the fore a strange phenomenon that they called 'ajar source'—they pointed out that many companies started releasing code,